

Statistical Analysis of Airbnb Prices Across Major European Cities: Patterns in Weekday and Weekend Listings

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Abstract—This study investigates the determinants of Airbnb prices across ten major European cities using a dataset containing 51,707 listings collected from weekday and weekend observations. Both descriptive and inferential statistical methods are employed to analyze pricing behavior, room characteristics, host attributes, and location-based factors. Independent sample t-tests, Welch ANOVA tests, correlation analysis, and multiple linear regression models are applied to evaluate the significance and strength of relationships between variables. The findings reveal that city-level differences and property characteristics are the strongest determinants of price variation, while accessibility variables such as metro distance exhibit only weak overall effects. However, interaction regression results indicate that the impact of metro accessibility differs significantly across cities. The study provides a comparative overview of Airbnb market structures across European urban markets and highlights the importance of geographical heterogeneity in short-term rental pricing.

I. INTRODUCTION

A. Overview

Since the 19th century, revolutions in transport technology have transformed tourism into a pillar of the global economy. In the digital era, the accommodation sector has seen a major shift with the rise of platforms like Airbnb, which decentralized the market by allowing individual hosts to compete with traditional hotels. This study examines a dataset of 51,707 listings across ten major European cities to analyze pricing dynamics and property characteristics during both weekdays and weekends.

B. Motivation

The large scale of Airbnb's dataset offers insights into urban pricing strategies and traveler preferences. This study analyzes how factors like geographical location, host type, and property capacity drive prices. By examining different cities and separating weekday from weekend listings, we illustrate the key patterns of the European short-term rental market.

C. Research Questions

The main research questions are:

- What is the overall distribution of prices in the merged dataset?
- Is there a statistically significant difference between weekday and weekend prices?
- Does room type have a significant effect on price levels?

- Is there a significant relationship between distance to the city center and price?
- Does metro distance significantly affect price, and does this relationship vary across cities?
- Do guest satisfaction levels differ significantly across cities?
- Are there significant differences in price distributions across cities?

D. Contributions

This study contributes to the literature on short-term rental markets in several ways. First, it provides a comparative analysis of Airbnb pricing structures across ten major European cities using a large-scale merged dataset. Second, the study combines descriptive statistics with inferential and econometric methods in order to distinguish statistically significant relationships from practically meaningful effects. Third, by incorporating interaction terms between metro accessibility and city indicators, the analysis highlights the heterogeneous role of transportation infrastructure across different urban markets.

II. METHOD

A. Dataset

1) *Overview*: The study utilizes a dataset obtained from Kaggle website [1] from ten cities: Amsterdam, Athens, Barcelona, Berlin, Budapest, Lisbon, London, Paris, Rome, and Vienna. Separate files for weekdays and weekends were merged, with new variables for "city" and "day_type" added during the process. The final dataset reached a size of 51,707 observations.

2) *Key Attributes*: Table I outlines the primary variables analyzed in this report.

TABLE I
MAIN ATTRIBUTES USED IN THE ANALYSIS

Variable	Type / Meaning
realSum / price	Numeric, listing price
city	Categorical, city name
day_type	Categorical, weekdays or weekends
room_type	Categorical, room category
room_shared	Boolean, shared-room indicator
room_private	Boolean, private-room indicator
person_capacity	Numeric, maximum guest capacity
host_is_superhost	Boolean, superhost indicator
multi	Binary, multi-listing host indicator
biz	Binary, business listing indicator
cleanliness_rating	Numeric, cleanliness rating
guest_satisfaction_overall	Numeric, overall guest satisfaction rating
bedrooms	Numeric, bedroom count
dist	Numeric, distance to city center
metro_dist	Numeric, distance to nearest metro

Note: A *superhost* refers to a host recognized by Airbnb for maintaining high ratings, low cancellation rates, and consistent activity. The *business listing indicator* (*biz*) denotes whether a listing is likely operated commercially rather than by an individual host. The *multi-listing indicator* (*multi*) shows whether a host manages multiple listings.

B. Method for Answering Research Questions

To move beyond descriptive analysis, statistical and econometric methods are employed to formally evaluate the relationships observed in the dataset. The choice of method depends on the nature of each research question and the type of variables involved.

First, an independent two-sample t-test is used to examine whether there is a statistically significant difference between weekday and weekend prices.

Second, one-way ANOVA tests are applied to investigate the impact of categorical variables on price and satisfaction outcomes. In particular, room type is analyzed to determine whether mean prices differ significantly across categories. ANOVA is also used to assess whether guest satisfaction levels and price distributions vary across cities.

Third, Pearson correlation analysis is employed as an initial measure of linear association between continuous variables, such as distance to city center and price. This allows for a preliminary understanding of the strength and direction of linear relationships before conducting regression analysis.

Fourth, the baseline model is extended by incorporating interaction terms between metro distance and city indicators to further investigate whether the effect of metro distance on price varies across cities. This interaction specification allows the marginal effect of metro accessibility on price to differ across cities, capturing potential heterogeneity in urban structure and transportation value.

The extended model can be expressed as:

$$\log(\text{Price}_i) = \beta_0 + \beta_1 \text{MetroDist}_i + \beta_2 \text{City}_i + \beta_3 (\text{MetroDist}_i \times \text{City}_i) + \epsilon_i \quad (1)$$

The statistical significance of the interaction terms is evaluated using joint F-tests to determine whether the relationship

between metro distance and price significantly varies across cities.

Finally, a multiple linear regression model is constructed to assess the joint effect of multiple variables on price. Given the right-skewed distribution of prices observed in the descriptive analysis, a logarithmic transformation of the dependent variable is applied. The baseline regression model includes key explanatory variables such as room type, distance to city center, metro distance, person capacity, host characteristics, and city-level fixed effects in order to control for structural differences across markets which can be expressed as:

$$\begin{aligned} \log(\text{Price}_i) = & \beta_0 + \beta_1 \text{RoomType}_i + \beta_2 \text{DistCenter}_i \\ & + \beta_3 \text{MetroDist}_i + \beta_4 \text{Capacity}_i \\ & + \beta_5 \text{HostCharacteristics}_i + \gamma \text{City}_i + \epsilon_i \end{aligned} \quad (2)$$

Model performance and statistical significance are assessed using standard metrics, including p-values (with a significance threshold of 0.05), R-squared, and Root Mean Squared Error (RMSE).

III. RESULTS

A. Descriptive Analysis

The analysis relies on histograms for distribution, pie charts for proportions, and scatter plots for identifying relationships between variables. To account for the imbalance in listing counts between cities (e.g., London vs. Amsterdam), percentage-based scaling is used for cross-city comparisons to prevent sampling bias.

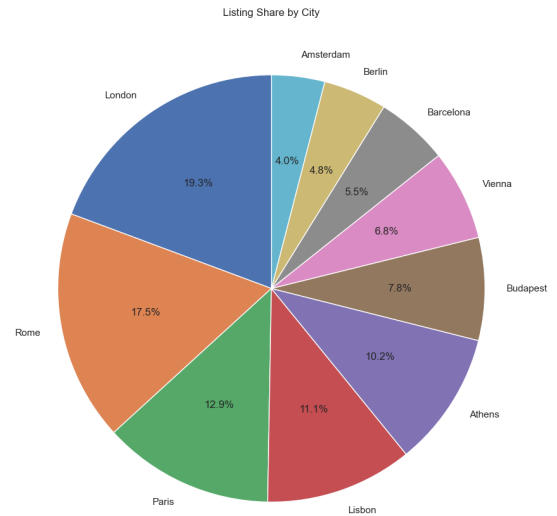


Fig. 1. Distribution of listings by city

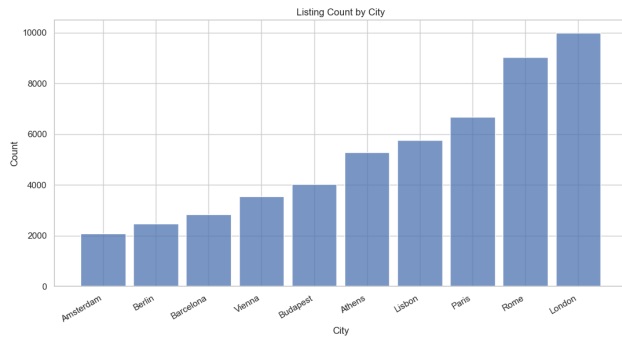


Fig. 2. Listing count by city

The merged data set contains 51,707 listings and is imbalanced between cities. London has 9,993 listings (19.33%), Rome 9,027 (17.46%), and Paris 6,688 (12.93%), while Amsterdam has 2,080 listings (4.02%). Therefore, city-level comparisons should be interpreted with this sampling structure in mind.



Fig. 3. Global price distribution (Log-scale)

Prices are strongly right-skewed. While the mean price is 279.88, the median stands significantly lower at 211.34. This gap, combined with a maximum value of 18,545, indicates a long upper tail driven by a luxury segment that pushes the average upward.



Fig. 4. Price variation by day type

As expected, weekends are generally more expensive. The weekend median price of 216.46 represents a slight premium over the weekday median of 207.15. While this difference confirms a weekend surge, the overlap between the two distributions is large.

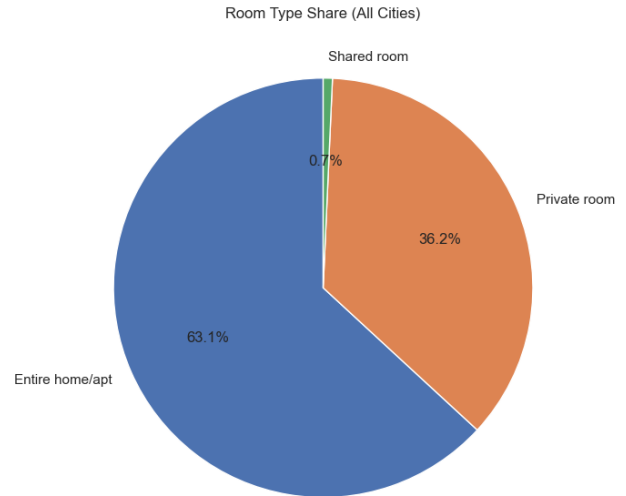


Fig. 5. Room type share

Room-type composition is dominated by Entire home/apt listings, followed by Private room, while Shared room has a small share.

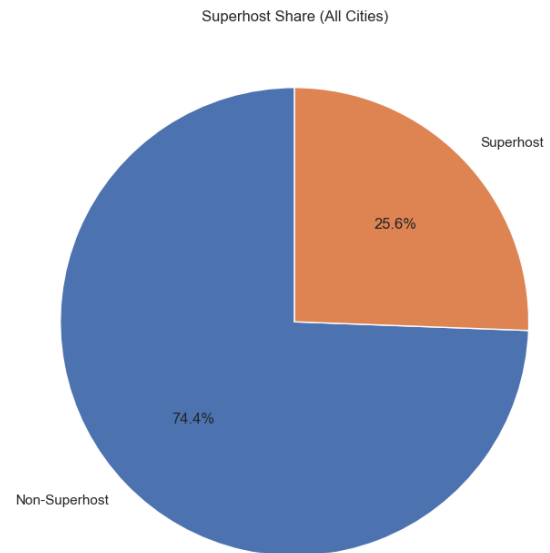


Fig. 6. Superhost Share

Superhosts represent 25.59% of the merged sample (non-superhosts: 74.41%).

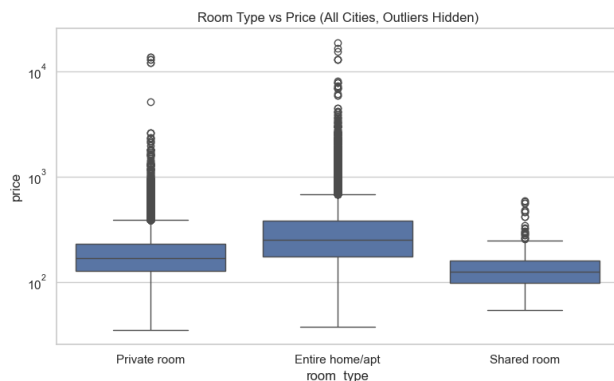


Fig. 7. Impact of room type on price

Room type strongly influences price. "Entire home/apt" listings have a median price of 248.46, while "Shared rooms" listings have a median price of 124.48. The large number of outliers in the "Entire home" and "Private room" categories shows that higher pricing is mostly concentrated in these room types, while shared rooms have relatively few outliers.

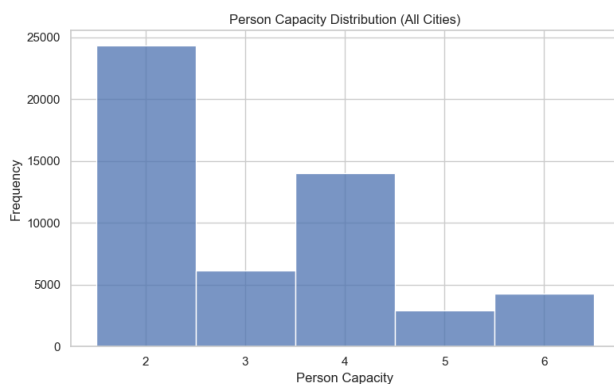


Fig. 8. Person capacity distribution

Capacity is concentrated in lower categories: 2-person listings = 24,333 (47.06%), 4-person listings = 14,000 (27.08%), 3-person listings = 6,165 (11.92%), 6-person listings = 4,274 (8.26%), and 5-person listings = 2,935 (5.67%).

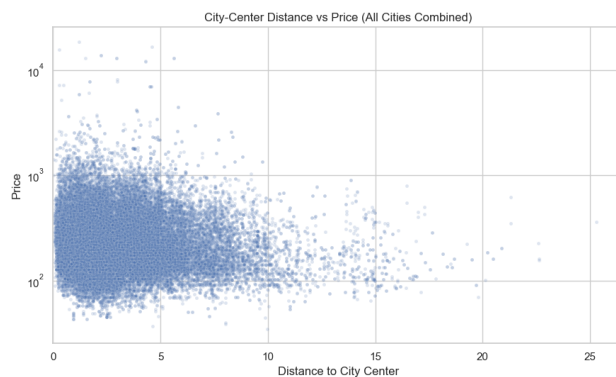


Fig. 9. Correlation between distance and price

The scatter plot suggests a very weak negative relationship between distance to the city center and price. The points are widely dispersed, and no clear linear pattern is visually apparent, indicating that distance alone does not strongly affect price variation in the dataset.

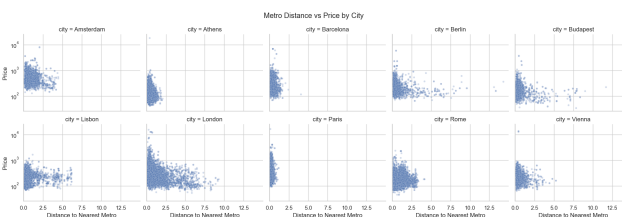


Fig. 10. Metro distance vs price by city

Similarly, proximity to the metro does not show a strong global effect. However, the slope intensity and point dispersion differ by city, so the metro-price relation is not uniform across markets.

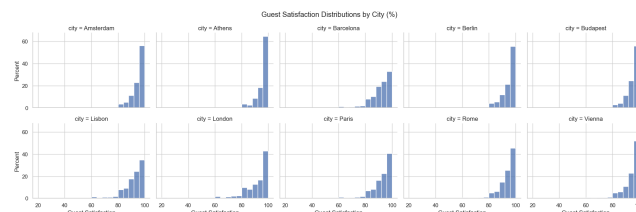


Fig. 11. Guest satisfaction across cities

Athens (95.00) and Budapest (94.59) lead in guest satisfaction, while London (90.65) records the lowest average ratings. Amsterdam stands out as the most expensive market (median 460.24), while Athens and Budapest present a "high-satisfaction, low-cost" profile.

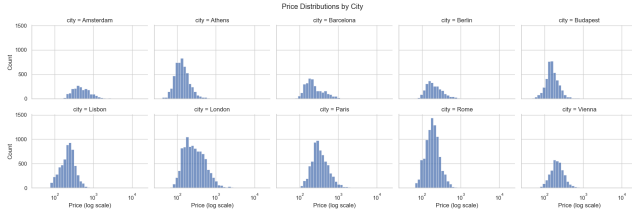


Fig. 12. Price distributions by city (log-scaled x-axis)

Median price is highest in Amsterdam (460.24), followed by Paris (317.60) and London (261.29), while it is lowest in Athens (127.72) and Budapest (152.98). This confirms meaningful cross-city variation in market structure.

B. Analysis for Answering Research Questions

This section presents the inferential statistical results used to answer the research questions formulated in Section 1.3.

1) Weekday vs Weekend Price Differences: To statistically evaluate whether there is a significant difference between weekday and weekend prices, an independent two-sample t-test is performed. The hypothesis testing framework is defined as follows:

- Null Hypothesis (H_0): There is no significant difference in mean prices between weekday and weekend listings ($\mu_{weekday} = \mu_{weekend}$).
- Alternative Hypothesis (H_1): There is a significant difference in mean prices between the two groups ($\mu_{weekday} \neq \mu_{weekend}$).

The test is conducted at a 95% confidence level ($\alpha = 0.05$). The decision is based on the p-value obtained from the t-test.

TABLE II
WEEKDAY VS WEEKEND PRICE COMPARISON AND T-TEST RESULTS

Metric	Weekday	Weekend
Sample Size (n)	25,500	26,207
Mean Price	275.68	283.96
Median Price	207.15	216.46
Std. Deviation	354.28	300.07
t-statistic	-2.8648	
Critical t-value	1.9600	
p-value	0.00417	
Levene Test p-value	0.291	
Cohen's d	0.0253	

The results indicate a statistically significant difference between weekday and weekend prices ($p < 0.05$). Therefore, the null hypothesis of equal means is **rejected**.

However, the magnitude of the difference is very small. The mean price difference is approximately 8 euros, and the effect size (Cohen's $d = 0.025$) suggests **negligible practical significance**. This implies that although weekend prices are slightly higher, the economic impact of this difference is **minimal**.

2) Effect of Room Type on Price: To examine whether room type has a statistically significant effect on price levels, a one-way ANOVA framework is applied. Since the assumption of equal variances is violated (Levene test, $p < 0.001$), the Welch ANOVA is used as an alternative.

The hypothesis framework is defined as:

- Null Hypothesis (H_0): All room types have equal mean prices.
- Alternative Hypothesis (H_1): At least one room type has a different mean price.

The test is performed at a 95% confidence level ($\alpha = 0.05$).

TABLE III
WELCH ANOVA RESULTS FOR ROOM TYPE AND PRICE

Metric	Value
Entire home/apt mean	324.315
Private room mean	204.940
Shared room mean	143.566
Levene test p-value:	1.71903×10^{-154}
F-statistic	1319.6072
p-value	1.34×10^{-302}
Eta-squared (η^2)	0.0317

The results indicate a statistically significant effect of room type on price ($p < 0.001$), leading to the rejection of the null hypothesis. However, the effect size ($\eta^2 = 0.0317$) suggests that room type explains only a relatively small proportion of the total variation in prices.

This implies that while room type affects the price, it is not the dominant determinant. Other factors, such as location, city-level effects, and host characteristics, likely play a more important role in price variation.

3) Effect of City on Price: To examine whether prices differ significantly across cities, a one-way ANOVA framework is employed. Since the homogeneity of variance assumption is violated (Levene test, $p < 0.001$), Welch ANOVA is used as a robust alternative.

The hypothesis framework is defined as:

- Null Hypothesis (H_0): All cities have equal mean prices.
- Alternative Hypothesis (H_1): At least one city has a different mean price.

The test is conducted at a 95% confidence level ($\alpha = 0.05$).

TABLE IV
WELCH ANOVA RESULTS FOR CITY AND PRICE

Metric	Value
F-statistic	592.4704
p-value	$< 1 \times 10^{-300}$
Eta-squared (η^2)	0.0954

The results reveal a statistically significant effect of city on price ($p < 0.001$), leading to the rejection of the null hypothesis. The relatively large F-statistic indicates substantial variation in mean prices across cities.

The effect size ($\eta^2 = 0.0954$) suggests that approximately 9.5% of the total variation in prices can be attributed to city-level differences. This is notably higher than the effect observed for room type, highlighting the importance of geographical location in determining Airbnb prices.

4) **Effect of City on Guest Satisfaction:** To analyze whether guest satisfaction differs significantly across cities, a one-way ANOVA framework is applied. Given that the assumption of equal variances is violated (Levene test, $p < 0.001$), Welch ANOVA is used.

The hypothesis framework is defined as:

- Null Hypothesis (H_0): All cities have equal mean guest satisfaction levels.
- Alternative Hypothesis (H_1): At least one city has a different mean satisfaction level.

The test is conducted at a 95% confidence level ($\alpha = 0.05$).

TABLE V
WELCH ANOVA RESULTS FOR CITY AND GUEST SATISFACTION

Metric	Value
F-statistic	185.8114
p-value	$< 1 \times 10^{-300}$
Eta-squared (η^2)	0.0309

The results indicate a statistically significant effect of city on guest satisfaction ($p < 0.001$), leading to the rejection of the null hypothesis.

However, the effect size ($\eta^2 = 0.0309$) is relatively small, suggesting that city explains only a limited portion of the variation in satisfaction scores. This implies that guest satisfaction may be more influenced by listing-specific factors such as cleanliness, host behavior, and service quality.

5) **Relationship Between Distance and Price:** To evaluate the relationship between distance to the city center and price, Pearson correlation analysis is conducted.

The hypothesis framework is defined as:

- Null Hypothesis (H_0): There is no linear relationship between distance and price ($\rho = 0$).
- Alternative Hypothesis (H_1): There is a linear relationship between distance and price ($\rho \neq 0$).

The test is performed at a 95% confidence level ($\alpha = 0.05$).

TABLE VI
CORRELATION RESULTS BETWEEN DISTANCE AND PRICE

Metric	Value
Pearson correlation (r)	-0.0447
R^2	0.0020
p-value	2.563×10^{-24}

The results indicate a statistically significant relationship between distance and price ($p < 0.001$), leading to the rejection of the null hypothesis. However, the correlation coefficient is extremely small ($r = -0.0447$), indicating a very weak negative linear relationship.

The coefficient of determination ($R^2 = 0.002$) shows that distance explains only 0.2% of the total variation in prices, suggesting it is a negligible factor.

This finding implies that although distance has a statistically detectable effect due to the large sample size, its practical impact on price is minimal. Therefore, distance alone is not a strong determinant of Airbnb prices, and other variables are likely to play a more important role.

6) **Relationship Between Metro Distance and Price:** To evaluate the linear relationship between metro distance and price, the Pearson correlation coefficient is computed.

The hypothesis framework is defined as:

- Null Hypothesis (H_0): There is no linear relationship between metro distance and price ($\rho = 0$).
- Alternative Hypothesis (H_1): There is a linear relationship between the two variables ($\rho \neq 0$).

The test is conducted at a 95% confidence level ($\alpha = 0.05$).

TABLE VII
PEARSON CORRELATION RESULTS FOR METRO DISTANCE AND PRICE

Metric	Value
Pearson correlation (r)	-0.0610
R-squared (R^2)	0.0037
p-value	7.376×10^{-44}

The results indicate a statistically significant negative relationship between metro distance and price ($p < 0.05$). Therefore, the null hypothesis is **rejected**.

However, the magnitude of the correlation is extremely weak ($r = -0.061$), and effect on price is negligible ($R^2 = 0.0037$). This implies that metro distance alone has almost no practical impact on price variation.

7) **Variation of Impact of Metro Distance on Price Across Cities:** To examine whether the effect of metro distance on Airbnb prices differs across cities, an interaction regression model between metro distance and city indicators is estimated. The interaction terms allow the effect of metro distance on price to vary across cities rather than assuming a constant relationship for the entire dataset. The joint significance of the interaction terms is evaluated using an F-test.

The hypothesis framework is defined as:

- Null Hypothesis (H_0): Interaction terms are jointly zero (the effect of metro distance is the same across all cities).
- Alternative Hypothesis (H_1): At least one interaction term is non-zero (the effect varies across cities).

The test is conducted at a 95% confidence level ($\alpha = 0.05$) with Berlin selected as the reference category.

TABLE VIII
JOINT F-TEST FOR METRO DISTANCE AND CITY INTERACTION

Metric	Value
F-statistic	96.7069
p-value	5.306×10^{-180}
R^2	0.3113
Adjusted R^2	0.3110

The results strongly reject the null hypothesis ($p < 0.05$), indicating that the effect of metro distance on Airbnb prices is not constant across cities.

The baseline coefficient of metro distance is negative and statistically significant ($\beta = -0.1455$, $p < 0.001$), suggesting that in the reference city, listings located farther from metro stations tend to have lower prices. However, the interaction terms reveal substantial heterogeneity across cities.

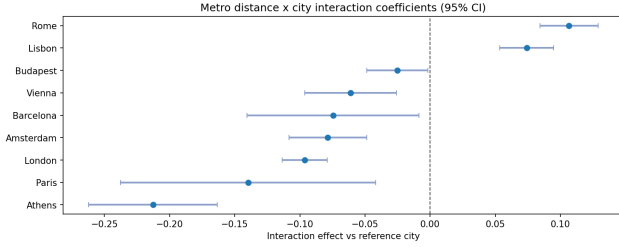


Fig. 13. Interaction coefficients of metro distance across cities

Figure 13 visualizes the estimated interaction coefficients for each city relative to the reference category. Positive coefficients indicate that the negative baseline effect of metro distance becomes weaker, while negative coefficients imply a stronger dependence on metro accessibility.

Among the interaction effects, Rome ($\beta = 0.1850$, $p < 0.001$) and Lisbon ($\beta = 0.1527$, $p < 0.001$) exhibit strong positive interaction coefficients, indicating that the negative baseline effect of metro distance becomes considerably weaker in these cities. In contrast, Athens shows a significant negative interaction effect ($\beta = -0.1342$, $p < 0.001$), suggesting that metro accessibility has a stronger impact on prices compared to the reference city. Significant positive interaction effects are also observed for Berlin and Budapest.

On the other hand, interaction terms for cities such as Barcelona, Vienna, London, and Paris are not statistically significant at the 5% significance level, implying that the relationship between metro distance and price in these cities does not differ substantially from the reference category.

Overall, these findings suggest that the economic value of metro accessibility varies significantly across urban markets. This may be related to differences in metro network coverage and the relative importance of public transportation in daily urban mobility.

8) Multiple Linear Regression and Key Drivers of Price: To identify the main determinants of listing prices, a multiple linear regression model is estimated using the logarithm of price as the dependent variable. The model includes structural, locational, host-related, and city-specific variables. Categorical variables for city and room type are included using dummy encoding, with Berlin selected as the reference category.

TABLE IX
REGRESSION MODEL SUMMARY

Metric	Value
Number of observations (n)	51,707
R-squared (R^2)	0.6131
Adjusted R-squared	0.6130

The model is statistically significant overall, and the null hypothesis that all coefficients are jointly zero is **rejected**. The model explains approximately 61% of the variation in log-transformed prices, indicating strong explanatory power.

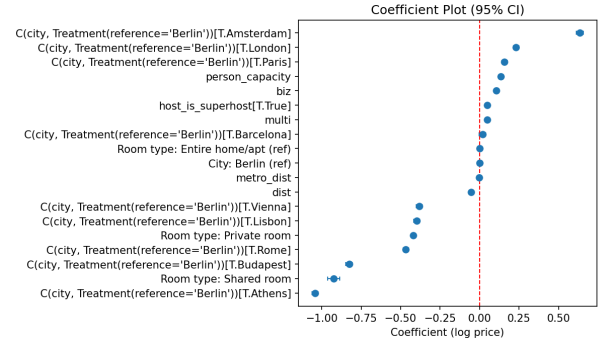


Fig. 14. Coefficient plot with 95% confidence intervals

The coefficient estimates show that **city-level effects are the strongest determinants of price variation**. Since Berlin is used as the reference category, all city coefficients are interpreted relative to Berlin.

Cities such as Amsterdam (+88.5%), London (+26.0%), and Paris (+17.2%) exhibit significantly higher prices compared to Berlin, while cities like Athens (-64.7%), Budapest (-56.3%), Rome (-37.4%), Lisbon (-32.9%), and Vienna (-31.8%) show substantially lower price levels.

Room type also has a strong effect. Compared to entire homes/apartments (reference category), *shared rooms* are approximately 60% cheaper and *private rooms* are about 34% cheaper.

Among continuous variables, **person capacity** has a strong positive effect, increasing prices by approximately 14.6% per additional guest. In contrast, **distance to the city center** has a modest negative effect (around 5.4% per unit increase), while **metro distance does not exhibit a strong global effect once other variables are controlled for**.

Host-related features such as being a superhost and operating a business listing have small but positive effects, increasing prices by approximately 5% and 11%, respectively.

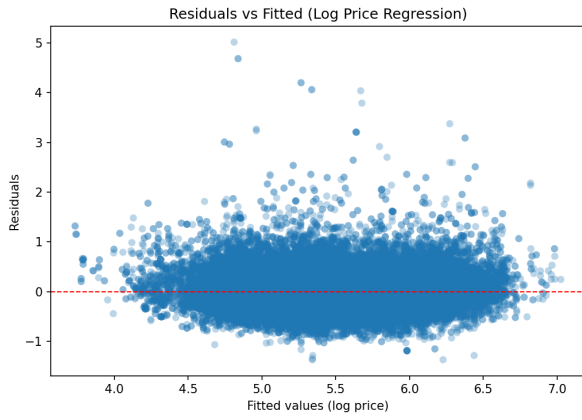


Fig. 15. Residual plot of the regression model

The residual plot indicates that the model provides a generally good fit, with no strong systematic pattern in residuals. However, some dispersion remains, suggesting the presence of unobserved factors and potential heteroskedasticity.

Overall, the findings suggest that **price formation in Airbnb markets is primarily driven by city-level differences and property characteristics**, while accessibility variables such as metro distance have limited explanatory power.

IV. CONCLUSION

This study examined Airbnb pricing dynamics across ten major European cities using a dataset containing more than 51,000 listings. Both descriptive and inferential statistical methods were employed to investigate the effects of room characteristics, location variables, host attributes, and city-level differences on listing prices.

The results show that while several variables are statistically significant, their practical importance differs considerably. In particular, room type, person capacity, and city-level effects emerge as the strongest determinants of price variation. Cities such as Amsterdam, Paris, and London display substantially higher price levels compared to other markets, confirming the importance of geographical and economic differences across cities.

Although distance to the city center and metro accessibility exhibit statistically significant relationships with price, their overall explanatory power remains relatively weak. However, interaction regression results reveal that the effect of metro accessibility varies significantly across cities, suggesting that transportation infrastructure plays different roles depending on urban context.

Overall, the findings indicate that Airbnb price formation is shaped primarily by structural and geographical factors rather than by accessibility variables alone. Future studies may extend this analysis by incorporating temporal dynamics, seasonal effects, neighborhood-level variables, or machine learning models to improve predictive performance and capture nonlinear relationships.

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- [1] Gyódi, K. (2023). "Airbnb Prices in European Cities", Kaggle. Available: <https://www.kaggle.com/datasets/thedevastator/airbnb-prices-in-european-cities>